

Technology Stack

Date	04 July 2026
Team ID	
Project Name	HDI Predictor Using Machine Learning
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3	Used to build a simple, responsive, and user-friendly web interface for entering HDI prediction inputs and displaying results.
2	Backend / Server-Side	Python 3.10, Flask	Flask provides a lightweight web framework that integrates easily with the machine learning model and processes prediction requests efficiently.
3	Database / Data Storage	HDI.csv Dataset, Pickle (.pkl)	The HDI dataset is used for model training, while the trained model is stored as an

			HDI.pkl file for fast prediction without retraining.
4	Cloud / Hosting / Deployment	Local Flask Server	The application is deployed locally for demonstration and testing purposes, making development and debugging simple and efficient.
5	Version Control & CI/CD	Git, GitHub	Git tracks source code changes, and GitHub provides version control, collaboration, and project repository hosting.
6	Third-Party APIs / Other Tools	NumPy, Pandas, Matplotlib, Scikit-learn, Jupyter Notebook, VS Code	Pandas and NumPy are used for data preprocessing, Matplotlib for visualization, Scikit-learn for model training, Jupyter Notebook for experimentation, and VS Code for development.